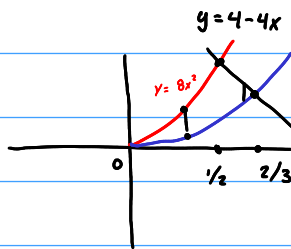


Section 6.1 Area Between Curves

85) Find the area in QI that is bound by the given curves $y = 3x^2$, $y = 8x^2$, $4x + y = 4$.

Quadrant 1

$$y = 4 - 4x$$



$$3x^2 = 4 - 4x$$

$$3x^2 + 4x - 4 = 0$$

$$(3x - 2)(x + 2) = 0$$

↓

$$x = \frac{2}{3}$$

$$8x^2 = 4 - 4x$$

$$8x^2 + 4x - 4 = 0$$

$$4(2x^2 + x - 1) = 0$$

$$4(2x - 1)(x + 1) = 0$$

↓

$$x = \frac{1}{2}$$

$$\int_0^{1/2} (8x^2 - 3x^2) dx + \int_{1/2}^{2/3} (4 - 4x - 3x^2) dx$$

$$\int_0^{1/2} \frac{5x^{2+1}}{2+1} dx +$$

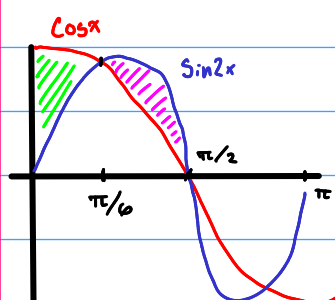
$$\left[\frac{5x^3}{3} \right]_0^{1/2} + \left[4x - 2x^2 - x^3 \right]_{1/2}^{2/3}$$

$$\frac{5\left(\frac{1}{2}\right)^3}{3} + \left(4\left(\frac{2}{3}\right) - 2\left(\frac{2}{3}\right)^2 - \left(\frac{2}{3}\right)^3\right) - \left(4\left(\frac{1}{2}\right) - 2\left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^3\right)$$

$$\frac{5}{8} \cdot \frac{1}{3} + \frac{8}{3} - \frac{8}{9} - \frac{8}{27} - 2 + \frac{1}{2} + \frac{1}{8}$$

$$= \boxed{0.315}$$

86) Find the total area between $y = \sin 2x$ and $y = \cos x$ over the interval $[0, \pi]$.



Since it's symmetric, we can integrate 0 to $\pi/2$ $\times 2$

$$\sin 2x = \cos x$$

$$2 \sin x \cos x = \cos x$$

$$2 \sin x \cos x - \cos x = 0$$

$$\cos x (2 \sin x - 1) = 0$$

↓

↓

$$\cos x = 0 \quad \sin x = \frac{1}{2}$$

$$x = \frac{\pi}{2} \quad x = \frac{\pi}{6}$$

$$2 \times \left[\int_0^{\pi/6} (\cos x - \sin 2x) dx + \int_{\pi/6}^{\pi/2} (\sin 2x - \cos x) dx \right]$$

$$2 \left(\left[\sin x + \frac{\cos 2x}{2} \right]_0^{\pi/6} + \left[-\frac{\cos 2x}{2} - \sin x \right]_{\pi/6}^{\pi/2} \right)$$

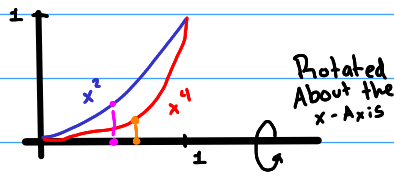
$$2 \left(\left[\sin \frac{\pi}{6} + \frac{\cos \frac{\pi}{3}}{2} - \sin 0 - \frac{\cos 0}{2} \right] + \left[-\frac{\cos \pi}{2} - \sin \frac{\pi}{2} + \frac{\cos \frac{\pi}{3}}{2} - \sin \frac{\pi}{6} \right] \right)$$

into the calculator...

$$= \boxed{1}$$

Section 6.2 Volumes

87) Consider the finite region in QI that is bound by $y = x^2$ and $y = x^4$. Find the volume if that region were rotated about the x-axis.



$$V = \int \pi r^2 dx$$

$$2 \cdot \int_0^1 \pi \left((x^2)^2 - (x^4)^2 \right) dx$$

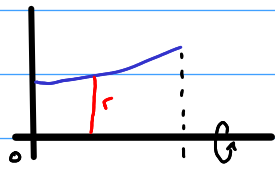
$$2\pi \int_0^1 (x^4 - x^8) dx$$

$$2\pi \left[\frac{x^5}{5} - \frac{x^9}{9} \right]_0^1$$

$$2\pi \left[\frac{1}{5} - \frac{1}{9} \right]$$

$$2\pi \cdot \frac{4}{45} = \boxed{\frac{8\pi}{45}}$$

88) Find the volume created when the region defined by $y = 1 + \sec x$ and $x = 0$ on the interval $[0, 1]$ is rotated about the x-axis.



Again $V = \int \pi r^2 dx$

$$\int_0^1 \pi (1 + \sec x)^2 dx$$

$$\pi \int_0^1 (1 + 2\sec x + \sec^2 x) dx$$

$$\pi \left[x + 2 \ln |\sec x + \tan x| + \tan x \right]_0^1$$

$$\pi \left[1 + 2 \ln |\sec 1 + \tan 1| + \tan 1 - \left(0 - \underbrace{2 \ln |\sec 0 + \tan 0|}_{2 \ln(1)} + \underbrace{\tan 0}_0 \right) \right]$$

plugged into a calculator...

$$= \boxed{15.739}$$